Đinh Trung Quốc Anh

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RESEARCH INTERESTS

Star Formation History of Massive Galaxies, Dusty Star-forming Galaxies and Galaxy Morphology at High Redshifts.

EDUCATION

International University, Vietnam National University - Ho Chi Minh City Vietnam
B.Eng in Space Engineering Sep, 2022 - Oct, 2026
All academic activities are in English
Cumulative GPA: 92.5/100 (3.87/4.00)
Thesis Title: Star Formation History of Massive Galaxies at z > 3, Using Non-parametric & Parametric Models
Advisor: Dr. Abdurro'uf (Indiana University)

RESEARCH EXPERIENCE

Independent Research Project: Spatially Resolved Star Formation History of Dusty Merging Galaxies at z > 2 from JWST+ALMA
Advisor: Dr. Abdurro'uf (Indiana University) Aug, 2026 - Present

- **Goal**: Understanding the impact of "wet" mergers on the star formation history (SFH) at Cosmic Noon.
- Adopted the close-pair detection and visual inspection to make a sample of 6 merging galaxies at z > 2.
- Conducted AGN subtraction by comparing the performances of galaxy-only model fitting and AGN model fitting on the central point-source's SED.
- Used piXedfit for pixel binning & Bagpipes/Dense Basis for fitting spatially resolved SED.
- Analyzed JWST/NIRSpec + the dust continuum from ALMA to investigate the interplay of dust & gas in these extreme systems.

2026 Summer Internship: Morphological Properties of Galaxies in a z = 4.53 Extended Galaxy Structure
Advisors: Prof. Arianna S. Long & PhD Student Nguyễn Bình (University of Washington) Jul, 2026 - Present

- **Goal**: Quantifying the morphology of over 200 galaxies in an extended galaxy structure at z = 4.53.
- Used statmorph to derive non-parametric and parametric morphological parameters in F277W + F444W filters.
- Analyzed the morphological relations in the extended galaxy structure such as the Gini-M20, the size-mass, the axis ratio - Sérsic index - effective radius relations.
- Classified the galaxies into early-, late-, merger- types using both Gini-M20 and CAS systems.

Bachelor Thesis: Star Formation History of Massive Galaxies at z > 3, Using Non-parametric & Parametric Models
Advisor: Dr. Abdurro'uf (Indiana University) Jan, 2026 - Jun, 2026

- **Goal**: Constraining the star formation history (SFH) of a sample of 80 massive galaxies at z > 3 and comparing the effects of non-parametric/parametric SFH models.
- Used Bagpipes for SED fitting, with 3 SFH models (Delayed Exponential, Double-Power Law, Non-parametric using Continuity priors).
- Compared the effects of 3 models on the SFH shapes and the look-back times at 25%, 50% and 75% of mass formed.
- Detected a subsample of 18 massive quiescent galaxies and 4 rejuvenating candidates based on their non-parametric SFHs and derived their formation, quenching, rejuvenating timescales.

Final-year Research Project: Measuring Brightness & Fitting the Spectral Energy Distribution of a Lensed Galaxy at z = 3.29
Advisor: Dr. Phạm Tuấn Anh (Vietnam National Space Center) Sep, 2025 - Jan, 2026

- **Goal**: Measuring the flux and SED-fit properties of a gravitationally lensed, high-redshift, Lyman-alpha emitting galaxy P198.

- Used SAOImageDS9 and Photutils to measure photometry of HST and JWST images.
- Conducted SED fitting using CIGALE and examined 2 SFH models: sfhdelayed and sfh2exp.

Third-year Research Project: Exploring the Interactions between Martian Crustal Magnetic Field and Solar Winds with MAVEN/SWEA
Advisor: Dr. Jaeheung Park (Korea Astronomy and Space Science Institute) Dec, 2024 - Jun, 2025

- **Goal**: Visualizing the altitude-resolved martian magnetic field's 2D topology and its correlation with the inward/outward flows of solar-wind electrons/photoelectrons.
- Calculated the shape parameter for topology classification with MAVEN/SWEA data.
- Identified the direction of electron populations with pitch angle & magnetic field's radial component.

2024 Summer Internship: Statistical Analysis of Ionospheric Irregularity Distribution Using Swarm Data
Advisors: Dr. Wookyoung Lee (Korea Astronomy and Space Science Institute) & Dr. Hyosub Kil (Johns Hopkins University) Jul, 2024 - Sep, 2024

- **Goal**: Identifying the scintillations at the high-latitude regions during summer of each hemisphere.
- Calculated and defined threshold for S parameter - an indicator for ionospheric irregularity.
- Plotted daytime/nighttime distributions of S parameter and plasma density in a year using Swarm LP data.

2023 Summer Internship: Front-end Testing and Deployment of BURSTT
Advisor: Dr. Chao-Te Li (The Academia Sinica Institute of Astronomy and Astrophysics) Jul, 2023 - Sep, 2023

- **Goal**: Understanding the front-end design for BURSTT radio telescope to detect Fast Radio Bursts.
- Assembled the Low-Noise Amplifier & Deployed an 2x16 antenna array at Fushan site with BURSTT team.
- Modelled a simple antenna model with Ansys-HFSS.

OBSERVING EXPERIENCE

Quantifying the Rotation of Saturn with Doppler Spectroscopy
Advisor: Dr. Phan Thanh Hiền (University of Science & Technology of Hanoi) Quy Nhon Observatory, Oct 2026

- Observed the tilted absorption lines of Saturn's spectrum with a 60-cm telescope at Quy Nhon Observatory.
- Reduced the spectrocopic data with Python and Demetra software; Measured the rotational velocity of Saturn by the Doppler shift of the H-alpha line.

Multicolor Observations of Globular Cluster NGC 2419 Using The 80-cm Twin Telescope (Teide Observatory, Spain)
Advisor: Prof. Phan Bảo Ngọc (International University, VNU-HCMC) Remote Observing, Dec 2024

- Submitted an observing proposal for NGC 2419 in the SDSS bands u', g', r', i', z'.
- Plotted the color-color, color-magnitude diagrams for NGC 2419.

PUBLICATIONS

1. *A city at the edge of time: Spectroscopic confirmation of a $z = 4.53$ extended galaxy structure.*
N. Bình, A.S. Long, R. Oh, **D.T.Q. Anh,** et al. 2026, in-prep
2. *Spatially resolved star formation history of dusty merging galaxies at $z > 2$ from JWST+ALMA.*
D.T.Q. Anh, Abdurro'uf. 2026, in-prep

HONORS & AWARDS

Gold Medal for GPA achievement, International University 2026
Full Scholarship, International University 2022 - 2026
Development Fund's Scholarship, Vietnam National University - Ho Chi Minh City 2022 - 2026
Dean's List, International University 2023, 2025
Silver Honor, International Astronomy and Astrophysics Competition 2025
Valedictorian of Entrance Exam, International University 2022
Certificate of Merit for College/University Valedictorians, Ho Chi Minh City People's Committee 2022

SKILLS

- **Programming:** PYTHON, C, MATLAB, SWIFT, SQL, HTML, CSS
- **SED-Fitting:** Integrated & Spatially Resolved SED fitting using Bagpipes, CIGALE, piXedfit, Dense-basis
- **Photometry Tools:** APT, Siril, SAO ImageDS9, Photutils, SEP
- **Galaxy Morphology Codes:** statmorph
- **Machine Learning Algorithms:** Random Forest, K-Means, SVM, DBScan, CNN
- **Antenna/Radio Simulations:** Ansys-HFSS, GNU-Radio

PARTICIPATED ASTRONOMICAL SCHOOLS

48th International School for Young Astronomers Sep 28 - Oct 16, 2026
Organized by The International Astronomical Union (IAU) & ICISE Gia Lai, Vietnam

SOKENDAI NAOJ Asian Winter School Feb, 2025 & Feb, 2026
Organized by The Graduate Institute for Advanced Studies - SOKENDAI Online

3rd TNU/SAGI Summer School for Observational Astrophysics Jul 27 - Aug 10, 2025
Organized by Simons Astrophysics Group at ICISE (SAGI) & Tây Nguyên University (TNU) Đăk Lăk, Vietnam

SERVICE

Member of Vietnam Amateur Astronomy Network (VAAN)	Aug, 2025 - Present
Class Monitor of K2022, Space Engineering	2022 - 2026
Member of Space Engineering's Admissions Counseling Team	2023 - 2026
Content Creator for 'Preparation for Entrance Exam' Program, Thanh Niên's Newspaper	2023, 2024, 2026
Head of Space Engineering's Astronomy Club (SEAC)	2023 - 2025
Member of Space Engineering's English Speaking Club (SEEC)	2023 - 2025
Co-organizer of Mid-Autumn Moon Gazing Event for World Space Week 2025	Oct, 2025
Calculus Tutor for Space Engineering Students	Oct, 2025 - Dec, 2025
Leader of IDEA Team, IU Guitar Club	Jul, 2023 - Jul, 2024

TALKS & PRESENTATIONS

Non-parametric/parametric star formation histories of massive galaxies at z $>$ 3 Aug, 2026
For University of Washington & University of Manchester Research Group Meeting Online

Introduction to Astro-Imaging Nov, 2025
For Seminar Hosted by Space Engineering's Astronomy Club Ho Chi Minh City, Vietnam

REFERENCES

Dr. Abdurro'uf (Department of Astronomy, Indiana University)
Email: fnuabdur@iu.edu

Prof. Arianna S. Long (Department of Astronomy, University of Washington)
Email: aslong@uw.edu

Dr. Phạm Tuấn Anh (Department of Astrophysics, Vietnam National Space Center)
Email: ptanh@vnsc.org.vn

Dr. Phan Hiền Vũ (Department of Physics, International University)
Email: phvu@hcmiu.edu.vn